IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-125

1. Topic (12 words max)

Voice Agentic Smart Reader Application in Sinhala and Tamil for Visually Impaired People

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Visually impaired (WHO, 2023) Sinhala and Tamil speakers face significant barriers accessing printed materials, despite global advances in assistive technology. According to the WHO, over 1 billion people experience vision impairment worldwide, limiting access to education and information. In Sri Lanka and Tamil-speaking regions, lack of robust tools for Sinhala/Tamil OCR (Purushoth Velayuthan, 2025)and expressive Text to Speech (TTS) exacerbates this gap. Benchmarks show that common OCR engines (e.g., Tesseract, EasyOCR) yield high error rates on Sinhala/Tamil documents due to complex scripts, legacy fonts, varied layouts, and real-world capture distortions (Oladele, 2024). Meanwhile, TTS systems for these languages often lack naturalness and emotional nuance, reducing comprehension and engagement for long-form content.

Compounding this, visually impaired users struggle to align cameras properly for document capture. Studies demonstrate that audio/haptic guidance (Shwetha Rajaram, 2025) markedly improves framing success, but existing solutions rarely target Sinhala/Tamil text characteristics (e.g., detecting upside-down Sinhala script). Blur, low light, skew, and orientation errors further degrade OCR accuracy (Martinez, 2025), leading to user frustration. Additionally, navigating lengthy documents via voice (Desai, n.d.) without intelligent segmentation or summarization forces linear reading, hindering efficient information access.

Thus, the research problem is to create a voice-agentic reader tailored for Sinhala and Tamil visually impaired users that: (1) provides voice/haptic camera-alignment (Jingyi Xie, 2025) guidance handling blur, focus, and upside-down pages; (2) generates emotionally expressive, context-aware TTS in Sinhala; (3) segments and summarizes with Sinhala LLM for long documents (Dawar, 2025) on demand; and (4) segments and summarizes with Tamil LLM for long documents (Dawar, 2025) on demand. Addressing these challenges can democratize access to printed and digital Sinhala/Tamil content, improving education, literacy, and inclusion.

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6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The Smart Voice Reader System is an assistive technology designed to empower visually impaired Sinhala and Tamil users by transforming printed or digital text into an interactive, emotionally expressive audio experience. It combines page alignment guidance, expressive Text-to-Speech (TTS), personalized summarization, and a voice-based learning assistant. The system ensures that users not only hear the text read aloud but also understand its emotional tone and meaning. Through real-time voice and haptic feedback, users can correctly align documents for accurate scanning. Key concepts and subtopics are automatically identified and summarized based on user preferences in Sinhala and Tamil, allowing easy navigation through content.
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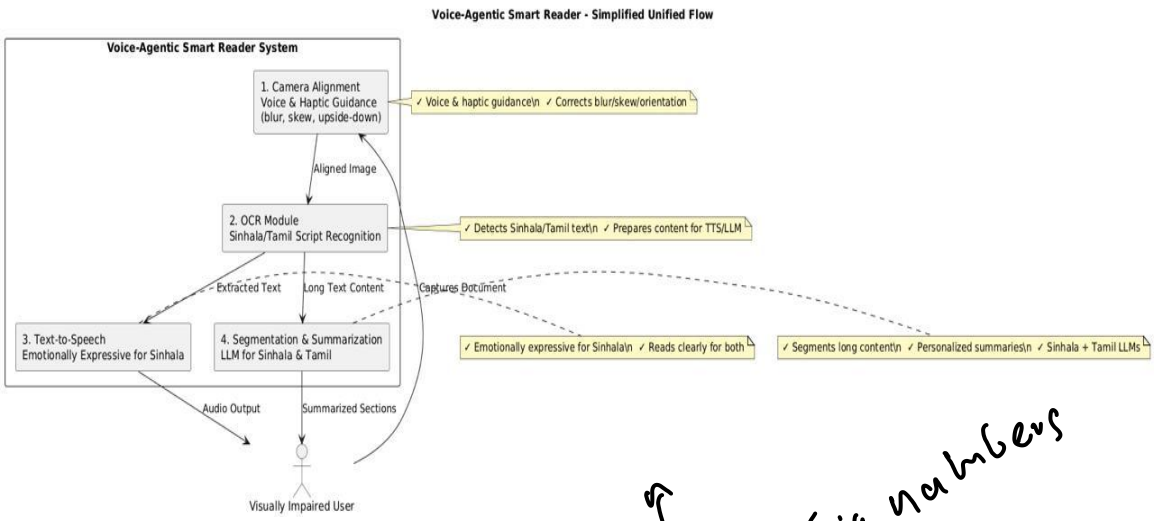

Have Fig numbers

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Domain Expertise - In Sri Lanka, about 1.7% of the population is blind, with the highest rates found in the Uva, Eastern, Northwestern, and Northern provinces. Cataracts and uncorrected refractive errors are the major causes, accounting for 90% of blindness cases. Braille devices, computer screen readers, and magnifiers are important advancements that assist blind individuals in understanding and reading written text.

Knowledge - NLP,Semantic understanding and entity recognition ,Sentiment and emotion detection from text, OCR, text to speech modeling,Linguistic phonetics ,Text segmentation, topic modeling, and summarization techniques ,User preference modeling and adaptive content generation,Dialogue management and conversational AI for voice interaction,Computer vision for document detection, alignment, rotation, and blur/focus detection,Multimodal feedback systems combining voice and haptics for user guidance,Knowledge base integration and semantic search for context-aware learning,Question answering, simplification algorithms, and educational technology for interactive learning,User modeling for personalized tutoring and follow-up conversations.

Data Requirements –

- Text of the books - Books must be available in digital formats for display on mobile devices.
- Audio files - The application should be capable of reading the text aloud using text-to-speech technology.
- Text-to-speech engine - A reliable TTS engine is needed on the device to provide clear and natural speech output.
- Book metadata - Details such as book title, author, and publication date are necessary to organize and categorize the books within the app.
- User preferences - The system should store user settings like font size and reading speed to offer a personalized reading experience.
- High-quality scanned images and digital texts in Sinhala and Tamil covering various document types (stories, instructions, academic texts, formal documents).
- Annotations highlighting key semantic elements (names, dates, places) for semantic-aware OCR training.
- Emotionally tagged text samples indicating tones like happy, sad, serious to train expressive TTS.
- Parallel speech datasets with expressive voices in Sinhala and Tamil for natural, context-aware speech synthesis.
- Structured documents with clearly defined subtopics to train subtopic detection and navigation.
- User interaction data including preferences on summary length, detail, and style for personalized summarization.
- Annotated summaries and segmented texts to train summarization models.
- Document metadata and user profiles to enable better content customization.
- Collections of Sinhala and Tamil document images captured under different real-world conditions (tilted, blurry, upside down) for page orientation correction and blur detection.
- Domain-specific knowledge bases (dictionaries, encyclopedias, educational materials) in Sinhala and Tamil for context-aware voice tutoring.
- Conversational datasets with typical user questions and follow-ups to enable interactive learning and clarifications.
- User interaction logs capturing questions, simplification requests, and quiz responses to improve the tutoring system.
- Speech and text corpora in Sinhala and Tamil for training natural, context-aware dialogue models powering the voice assistant.

8. Objectives and Novelty

Main Objective

The main objective of this research is to develop a voice-agentic smart reader application specifically designed to empower visually impaired Sinhala and Tamil speakers by integrating advanced, human-centered technologies that go beyond traditional assistive tools. The system will combine semantic-aware Optical Character Recognition (OCR) with emotionally expressive Text-to-Speech (TTS) to not only read text from complex Sinhala and Tamil documents but also understand the deeper meaning and convey it through natural, emotion-rich voice output. It will further incorporate intelligent subtopic detection and user-personalized summarization features, enabling non-linear navigation and tailored content consumption through simple voice commands, thus improving efficiency and user control. Additionally, the system will feature real-time haptic and voice-guided document alignment assistance to help users independently capture well-oriented document images, ensuring high OCR accuracy even under imperfect conditions. Finally, an interactive, context-aware voice agent will enrich the user's learning experience by proactively offering explanations, cross-referencing unfamiliar concepts with external knowledge bases, and supporting natural follow-up questions in Sinhala and Tamil. Together, these components aim to provide a seamless, intelligent, and empowering reading and learning experience tailored to the needs of visually impaired users in low-resource language communities.

Member Name with Registration No	Sub Objective	Tasks	Novelty
SIRIWARDANA B J A IT22336590	To design and implement a combined OCR and TTS system that can accurately extract Sinhala and Tamil text from diverse document formats and generate emotionally	• Sinhala Text Dataset Collection and Preprocessing - To build an emotionally expressive TTS system, the first step is collecting a large and diverse set of Sinhala text data from books, news articles, dialogues, and social media. These texts are then cleaned and preprocessed	This research component introduces two key novelties to advance accessible speech technology for the Sinhala language. First, it presents the development of an emotionally expressive Text-to-Speech (TTS) system tailored for Sinhala, enabling natural and emotion-

	expressive speech by understanding the semantic and emotional context of the content.	by removing unnecessary symbols, standardizing character encoding, and breaking the content into sentence-level units. This cleaned data is used to train both emotion recognition and TTS models, ensuring that the system understands the structure and flow of Sinhala language accurately. (F.A. Rizvi, 2025) • Motion and Sarcasm Detection - Emotion detection is performed using advanced Natural Language Processing (NLP) techniques. Transformer-based multilingual models like XLM-R or IndicBERT are fine-tuned to detect emotional cues in Sinhala sentences. Along with emotions, the system also detects sarcasm, which requires a nuanced understanding of both syntax and context. This helps distinguish between sincere and ironic expressions, which is important for	aware voice output for visually impaired users. This builds upon our previously completed work on Tamil emotional TTS, extending similar capabilities to another underrepresented South Asian language. (Jubeerathan Thevakumar, 2025) Second, it addresses a major limitation in current emotion recognition models by introducing a method to detect complex and mixed emotions within a single sentence, such as combinations of happiness and nervousness or sadness and hope. Together, these innovations enhance both the emotional depth and linguistic relevance of Sinhala speech synthesis, offering a more engaging and context-aware auditory experience. (Sherine Nagy Saleh, 2025) (Harinda Fernando, 2025)

		delivering accurate emotional tone in voice synthesis. (Mr. Suryavamshi Sandeep Babu, 2025) • Emotion Categorization into Defined Emotional Classes - After detecting emotional content, each sentence is categorized into one or more defined emotion classes such as happy, sad, angry, excited, worried, curious, sarcastic, or neutral. Multi-label classification methods are applied to account for sentences that contain overlapping emotional signals. These emotional tags guide the speech engine in selecting the appropriate expressive tone, thereby making the speech output more natural and contextually meaningful. (Farah Hassan Labib, 2025) • Sinhala Emotional TTS Model Development - The TTS engine is trained to synthesize Sinhala speech with expressive tone	

		using a dataset of recorded Sinhala sentences spoken with different emotions. Models like Tacotron 2 or Coqui TTS are customized to include emotion embeddings, which allow the system to control speech elements like pitch, speed, and intonation. As a result, the synthesized voice does not sound robotic, but instead mirrors the actual emotional intent of the content, making it suitable for visually impaired listeners who rely on vocal cues. (Senarath, 2024) • Context-Aware Voice-Agentic Interaction - To make the application interactive and intelligent, context-aware voice prompts are integrated. These prompts help engage the user in a natural flow of reading and interaction. For example, after reading a paragraph, the system can ask, “Would you like me to explain further?” or “Shall I continue to the next section?”.	

		These voice-agentic features help the user control their reading experience smoothly, without needing any visual input. (Dr. B. Esther Sunanda, 2024)	
HEWAGAMA A D IT22316486	To develop an intelligent content segmentation and summarization system that enables visually impaired Sinhala and Tamil users to navigate long-form documents through voice commands, access personalized summaries on demand, and enhance reading efficiency and comprehension.	• Preparation of the Sinhala Dataset - An effective Sinhala content processing system requires a diverse dataset, including newspapers, academic texts, books, and official documents. Preprocessing is essential to clean the data by removing HTML tags, punctuation issues, and formatting errors. For scanned or printed sources, OCR tools like Tesseract (with Sinhala support) should be used. To train the model on summarization and segmentation tasks, the dataset should be manually annotated with summaries and section boundaries.	This research component introduces two key novelties.First , Sinhala language model will be developed by combining content segmentation and personalized summarization using a Large Language Model (LLM) within a Retrieval-Augmented Generation (RAG) architecture (Sawiras, 2024). The system will intelligently break long Sinhala documents into meaningful sections and generate user-specific summaries based on individual preferences such as summary length, topic focus, and language simplicity. By integrating segmentation and personalization into the LLM workflow, the application aims to deliver a more accessible and tailored reading experience, especially useful for

		• Building or Fine-Tuning a Sinhala Language Model – Due to the limited availability of Sinhala-specific models, multilingual models like mT5, XLM-R, or mBART can be fine-tuned using the prepared dataset. These models support transfer learning and can be adapted using frameworks such as Hugging Face Transformers. For low-resource environments, efficient tuning methods like LoRA or QLoRA are recommended. This step enhances the model's ability to understand and generate Sinhala summaries and segmented outputs. (Hansaka & Sumanathilaka, 2025) (Ravindu Jayakody, 2024) • Setting Up the Retrieval System (RAG) - The RAG architecture combines retrieval and generation. Sinhala documents should be divided into meaningful chunks (e.g., paragraphs or sections), which	users navigating complex or lengthy texts through voice interaction. Existing systems have segmentation and summarization separately. Second, to make article segmentation more accurate and reliable, future work should focus on using smart, learning-based methods instead of just rule-based ones. For example, using deep learning can help better detect and remove images, ads, and other non-article parts from newspaper pages. Fixing problems like tilted pages (skew) and unclear boundaries can also improve results, especially with old or scanned newspapers. Adding language models that understand the meaning of the text can help separate article content from unrelated items. These improvements will make it easier to correctly find and summarize articles in different types of newspaper layouts. (Gayashan, Perera, Nirmani, & Ranathunga,

		are then converted into vectors using models like LaBSE or multilingual sentence-transformers. These vectors, along with metadata, are stored in vector databases such as FAISS or ChromaDB. This allows the system to fetch contextually relevant Sinhala content for user queries. (Senanayaka, 2025) • Construction of the RAG Pipeline - Once the retriever is in place, it is connected to the generator to form the complete RAG pipeline. Queries are converted into vectors, relevant content is retrieved, and the LLM generates a summary based on that context. This improves the accuracy and relevance of the summarization, as it incorporates real document data rather than relying solely on pre-trained knowledge. (Wickramasinghe, 2025)	Old Sinhala Newspaper Article Segmentation for Content Recognition Using Image Processing, 2022)

		• Integration of Voice and Navigation Features - Voice interaction is essential for user-friendly access, especially for visually impaired users. Speech recognition tools like Vosk or Mozilla DeepSpeech can process Sinhala voice commands. Text-to-speech systems such as Google Cloud TTS or Mozilla TTS enable audio responses. Content segmentation allows users to navigate non-linearly, accessing summaries or sections directly through voice commands. • Implementation of a Personalization Layer - To deliver tailored experiences, the system should include user profiles that store preferences such as summary length, language style (simple or formal Sinhala), and topic focus. These preferences are used to adjust the prompt sent to the language model, ensuring the generated summary aligns with individual	

		needs. This results in a more accessible and user-centric output. (M Jahan, 2023)	
ALUWIHARE M C IT22326454	Develop a robust, user-friendly system optimized for visually impaired users that enables accurate capture and reading of Sinhala language documents through real-time camera guidance, quality assurance, orientation detection, and precise OCR integrated with expressive TTS.	• Camera Alignment & Framing – ○ Guides the user to point the phone correctly using voice and vibration instructions (e.g., “move left,” “go closer”). ○ Check whether the document is fully inside the frame, centered, and correctly oriented (not upside down). • Focus and Blur Detection – ○ Ensures the image is not blurry before capturing by analyzing focus sharpness using image processing. ○ If the image is too blurry, gives an audio warning like “please hold steady” or “image is not clear.”	The proposed system’s core innovation is its language-specific focus on Sinhala script—an area largely underserved by mainstream mobile OCR tools such as Google Lens or Microsoft’s Seeing AI. By integrating a deep-learning OCR engine trained explicitly on Sinhala characters, the app achieves far higher recognition accuracy on complex ligatures and diacritics than generic models. To ensure the camera is correctly framed before capture, it provides real-time voice and haptic guidance: audio prompts (e.g. “move left,” “tilt up”) and subtle vibrations signal when the document edges are entering the camera viewfinder, drawing on techniques proven effective in mobile OCR aids for the blind. (Manduchi & Coughlan, 2014) (Sterence, Du, Ross, & Jou, 2016)

		• Document Orientation Detection –
		  ○ Detects if the document is upside down by analyzing the orientation of Sinhala and Tamil characters using deep learning or OCR direction confidence.
		  ○ Automatically rotates the captured image or instructs the user to flip the phone: "Rotate phone 180 degrees."
		• OCR for Sinhala Script –
		  ○ Once the document is captured and aligned, it extracts Sinhala and Tamil characters using OCR (Optical Character Recognition).
		  ○ Send the recognized Sinhala and Tamil text to the TTS module for reading aloud.
			Once alignment is satisfactory, the system performs pre-capture blur detection by analyzing image sharpness in real time. If motion or defocus is detected, the user hears a brief alert tone and is prompted to steady the device—an approach shown to significantly reduce failed OCR attempts due to blurred inputs. (Courtney, 2021)
			
			Unlike most smartphone readers, our app also includes automatic orientation correction: it detects when a page is held upside-down and either rotates the image programmatically or issues a clear spoken instruction (e.g. "rotate 180 degrees"), ensuring the text is right-side-up before recognition—a feature validated in recent accessible OCR studies. (Manduchi & Coughlan, 2014)
			
			By chaining these components—Sinhala-optimized OCR, dynamic alignment feedback, blur filtering, and orientation adjustment—the system delivers a seamless end-to-end reading flow: visually

			impaired Sinhala speakers can independently capture, recognize, and listen to printed text with confidence that every scan meets the quality standards necessary for accurate TTS rendering.
SHARUMATHAN M IT22348234	To develop an intelligent content segmentation and summarization system that enables visually impaired Tamil users to navigate long-form documents through voice commands, access personalized summaries on demand, and enhance reading efficiency and comprehension – optimized for on-device use.	• **Preparation of the Tamil Dataset** An effective Tamil content processing system requires a curated dataset that includes newspapers, academic documents, Wikipedia articles, and government communications. Sources such as CulturaX (Team, 2023) (a 12GB Tamil dataset), translated Alpaca (52K samples), and OpenOrca subsets (93K samples) are used. Preprocessing steps include cleaning HTML tags, fixing encoding, and segmenting data for summarization and segmentation tasks. Summaries and section boundaries are manually annotated to support both supervised training and evaluation. Instructional content is prioritized to suit the needs of visually impaired learners.	1.On-Device Tamil LLM Summarization: First to implement Tamil-specific summarization using quantized LLaMA models that run fully offline on mobile devices. This improves accessibility for rural and low-connectivity regions while preserving user privacy. (Dettmers T. P., 2023) 2.Instruction-Tuned Personalization for Visually Impaired Users: Built on Tamil-LLAMA, the system adapts to document type, voice preferences, and task (e.g., explain vs. summarize) using fine-tuned LoRA adapters. A real-time feedback loop ensures summaries evolve with user habits. (Abhinand, 2024)

		• **Building or Fine-Tuning a Tamil Language Model** Fine-tune the **Tamil-LLAMA 7B** or **Tamil distilled model** using **LoRA** and **QLoRA** for parameter-efficient training. Apply **4-bit quantization** and convert to **GGUF** format for use in llama.cpp, ensuring smooth performance on mid-range Android phones. Benchmark against traditional cloud-based summarizers for accuracy, latency, and memory usage. (Abhinand, 2024) The model leverages the Tamil-LLAMA project, which expands LLaMA 2's vocabulary with 16,000 Tamil tokens for better representation. Fine-tuning is conducted using Low-Rank Adaptation (LoRA) and quantization methods like QLoRA to enable deployment on low-resource devices. The model is further optimized for summarization using instruction tuning datasets. Deployment-ready versions are converted to GGUF format for compatibility with llama.cpp, enabling real-time inference on mobile phones. (Abhinand, 2024) (Feng, 2020)	

		• Setting Up the Retrieval System (RAG) The RAG (Retrieval-Augmented Generation) architecture enables dynamic, context-aware summarization. Tamil documents are split into meaningful discourse chunks (paragraphs, sections) and embedded into vector representations using LaBSE or multilingual MiniLM. These are stored in vector databases like ChromaDB or FAISS, which can be deployed locally. When a user asks a query via voice, the system retrieves the most relevant chunks and passes them to the language model to generate precise, contextual summaries. (Feng, 2020) **• Construction of the RAG Pipeline** The summarization pipeline integrates the retriever with the generator (Tamil LLaMA) to form a low-latency, fully on-device summarizer. Queries are converted into embeddings, relevant content is fetched, and then summaries are generated based on that context. Compared to traditional cloud-based systems, this pipeline allows real-time	

		offline access, which is essential for users with limited or no internet connectivity. **• Integration of Voice and Navigation Features** Voice interaction is critical for blind and visually impaired users. Tamil speech-to-text tools like Whisper or Vosk are integrated to process user commands, while Coqui TTS or Google Cloud TTS provides natural Tamil voice output. Voice-based navigation enables non-linear access to content, where users can request summaries of sections or ask the assistant to repeat, explain, or simplify paragraphs. This interaction happens entirely on-device to enhance privacy and responsiveness. (Gerganov, 2023) **• Implementation of a Personalization Layer** User profiles store preferences such as summary length, topic focus, tone (formal or conversational Tamil), and language simplicity. These preferences are used to customize prompts sent to the LLM during summary generation. Additionally, a feedback loop collects voice-based reactions and rating responses, helping	

		the model learn user habits and adapt future outputs accordingly. This ensures a more accessible, user-centered summarization experience. Offline Learning Assistant and Adaptive Summary **Engine**Incorporate **real-time feedback loop** where users can rate or react to summaries.Adjust summarization prompts or retrain lightweight adapters based on feedback trends.Enable follow-up voice queries like “simplify that,” “repeat section,” or “explain word.” (Dettmers 2024)	

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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
SIRIWARDANA B J A IT22336590	This component leverages natural language processing (NLP) and machine learning techniques to extract not only textual content but also semantic meaning from Sinhala and Tamil documents. Advanced sentiment analysis models detect emotional tones, enabling the text-to-speech system to generate expressive, context-aware narration. Data science methods in text analytics and speech synthesis play a key role in transforming raw OCR output into meaningful, emotionally rich audio content.
HEWAGAMA A D IT22316486	This component segments complex Sinhala documents into subtopics using topic modeling and clustering, enabling non-linear navigation. It enhances summarization by integrating Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) architecture to produce context-aware summaries, even in low-resource languages like Sinhala. Personalization is achieved through user interaction data and preferences, generating summaries that adapt to individual needs. The system leverages unsupervised learning, user behavior analytics, and natural language generation for efficient, intelligent, and user-focused content summarization.
ALUWIHARE M C IT22326454	Computer vision and sensor data analysis techniques are applied to detect document alignment and orientation in real time. Image processing algorithms evaluate blur and tilt, while accelerometer and gyroscope data support precise feedback via voice and haptics. Data-driven models provide instant, adaptive guidance, enhance OCR accuracy and enabling independent scanning by visually impaired users through multimodal feedback integration.
SHARUMATHAN M IT22348234	This component segments complex Tamil documents into subtopics using topic modeling and clustering, enabling non-linear navigation. It enhances summarization by integrating Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) architecture to produce context-aware summaries, even in low-resource languages like Tamil. Personalization is achieved through user interaction data and preferences, generating summaries that adapt to individual needs. The system leverages unsupervised learning, user behavior analytics, and natural language generation for efficient, intelligent, and user-focused content summarization.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr	Mahima	Weerasinghe	
Co-Supervisor	Dr	Lakmini	Abeywardhana	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Novelties should be streamlined further
Asses the "failure mode" & have a plan to face them

Staff Member's Name	Signature
Dr. Mahima Weerasinghe	

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.